

CAREEM AI CHALLENGE SUBMISSION

Challenge #2: The Auto-Analyst | Candidate: **Syed Taha Zaidi** | [GitHub Repo](#) | [LinkedIn](#) | [Portfolio](#)

IDEA & APPROACH

I built a zero-prompt Auto-Analyst that ingests Careem Care support-ticket data and autonomously produces an executive briefing — no human prompting required. The system runs six analysis modules: KPI computation, vertical drill-down, contact-reason pain-point detection, channel efficiency comparison, cross-market automation gap analysis, and weekly anomaly detection (z-score based). It scores each contact reason for automation potential using a composite metric (volume × dissatisfaction × simplicity × resolvability) and generates ranked recommendations with estimated agent-minutes saved. The output is a self-written narrative report with severity-tagged insights, three data-driven takeaways, and a single priority action point — ready for a VP of Care.

ARCHITECTURE

CSV Input → Data Loader → Analysis Engine (6 modules) → Report Generator → Executive Briefing

Analysis Modules:

1. KPI Computation — CSAT, FCR, Deflection Rate, Handle Time, Escalation Rate
2. Vertical Analysis — Food, Ride, Quik, Pay comparison with anomaly flagging
3. Contact Reason Pain-Point Detection — High-volume + low-CSAT identification
4. Channel Efficiency — In-App Chat vs Phone vs Email vs Social performance
5. Cross-Market Comparison — UAE, KSA, Pakistan, Jordan, Egypt automation gaps
6. Trend & Anomaly Detection — Week-over-week z-score spike/dip detection

Automation Scoring: $\text{score} = (\text{volume_wt} \times 30) + (\text{dissatisfaction} \times 25) + (\text{simplicity} \times 20) + (\text{resolvability} \times 25)$

SAMPLE OUTPUT (KEY METRICS FROM 5,000-TICKET ANALYSIS)

Avg CSAT:	3.64 / 5.00
First Contact Resolution:	79.8%
Deflection Rate:	16.7%
Avg Handle Time:	9.0 min
Escalation Rate:	12.4%
Repeat Contact Rate:	11.8%

AI-SURFACED INSIGHTS (AUTO-GENERATED)

[!] Top automation candidates: "Late delivery" (score 49.8, 382 tickets), "Missing items" (score 49.2, 349 tickets), "Transfer failed" (score 48.9, 106 tickets).

[!] Pay vertical FCR at 77.8% is below 80% target — escalation overhead increasing cost-to-serve.

[!] Channel gap: In-App Chat (4.9 min) vs Email (17.9 min) = 3.6x efficiency difference.

[!] Week 27 anomaly: dip to 29 tickets (z-score: -4.43). Requires investigation.

[+] Volume trending down (-12.6%) while CSAT improved (+0.09) — efficiency gains working.

PRIORITY ACTION POINT

Pilot AI-driven triage for "Late delivery" — automation score 49.8/100, 382 tickets, potential to save ~3,318 agent-minutes per period. This single intervention could improve deflection rate by ~7.6pp and reduce cost-to-serve measurably.

TECHNICAL DETAILS

Language: Python 3.8+ (stdlib only — zero external dependencies)
Dataset: Self-created, 5,000 tickets, 14 fields, modeled on Careem Care ops
Time to build: ~45 minutes
Lines of code: ~350 (analyst) + ~100 (data generator)
GitHub Repo: github.com/taha-zaidii/Careem-AI-Auto-Analyst-Challenge

Files submitted:

- auto_analyst.py — Core analysis engine
- generate_dataset.py — Dataset generator
- careem_care_tickets_h1_2025.csv — Sample data
- auto_analyst_report.txt — Full auto-generated report
- README.md — Documentation with architecture diagram

WHY THIS APPROACH MATTERS FOR CAREEM CARE

This tool directly addresses Careem Care's operational needs: it tracks the exact KPIs from the job description (contact rate, deflection rate, CSAT, FCR, automation accuracy), identifies automation opportunities aligned with the "Care-First Automation" philosophy, and produces actionable executive summaries — the kind of output an Automation & AI Manager would need to present to Care leadership weekly. It demonstrates the analytical mindset, Python proficiency, and ML-readiness (the scoring engine is a simple model that could easily be replaced with a trained classifier) that the role requires.